

Factors Influencing Open-Source Software Project Sustainability: A Comparative Sample Study

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Abstract—Open-source software (OSS) has become the foundation of modern software development, with 96% of codebases containing OSS components. However, a significant portion of these projects face sustainability challenges: 91% of codebases contain components with no development activity in the past two years. This study investigates factors that differentiate sustainable from non-sustainable OSS projects through a comparative sample study of 500 GitHub repositories (250 sustainable, 250 non-sustainable). We analyze governance practices, community health indicators, and ecosystem metrics to identify predictive factors. Our results show that maintainer guidelines (OR=4.22, $p<0.0001$), bus factor (median 4 vs. 2, $p<0.0001$), and stars threshold ($\geq 11,934 = 4.15\times$ lift) significantly predict sustainability. A combined XG-Boost model achieves 97.2% prediction accuracy (AUC=0.993). We conclude that governance maturity and contributor structure predict sustainability better than popularity metrics alone, providing actionable insights for maintainers, contributors, and organizations evaluating OSS dependencies.

Index Terms—open-source software, sustainability, GitHub, governance, machine learning

I. INTRODUCTION

A. Problem and Motivation

Open-source software (OSS) has become the foundation of modern software development. According to recent industry reports, 96% of codebases contain OSS components [?]. However, a significant portion of these projects face sustainability challenges: 91% of codebases contain components with no development activity in the past two years [?]. This creates substantial risk for organizations depending on OSS.

The problem matters for several reasons. First, security vulnerabilities in unmaintained projects remain unpatched, creating attack vectors. Second, breaking changes in dependencies cascade through ecosystems. Third, organizations invest in projects that may become abandoned, leading to costly migration efforts.

This problem affects multiple types of practitioners:

- **OSS maintainers** need guidance on governance practices that promote longevity
- **Contributors** must assess project health before investing effort
- **Organizations** need to evaluate dependency risks
- **Funders** require evidence-based criteria for supporting projects

The practical context appears when practitioners must decide which projects to depend on, contribute to, or fund, but lack clear indicators of long-term sustainability.

B. Research Questions

We address the following research questions:

- **RQ1:** What governance and documentation practices differentiate sustainable from non-sustainable projects?
- **RQ2:** How do community health indicators (response times, bus factor, contributor diversity) correlate with survival?
- **RQ3:** Do ecosystem metrics (stars, forks, watchers) correlate with sustainability?
- **RQ4:** Which factors best predict sustainability when combined?

C. Contributions

This paper makes the following contributions:

- 1) First study combining governance, community, and ecosystem dimensions in one predictive framework
- 2) Novel operationalization of bus factor and maintainer guidelines metrics
- 3) Interpretable ML model achieving 97.2% accuracy with practical thresholds for practitioners

II. RELATED WORK

A. OSS Sustainability

Crowston and Howison [?] established foundational metrics for OSS community structure, finding that successful projects exhibit specific communication patterns. Chengalur-Smith et al. [?] conducted a longitudinal study defining sustainability through project lifespan and activity patterns. Coelho and Valente [?] analyzed why modern open source projects fail, identifying common patterns of abandonment.

B. Governance and Documentation

Schweik and English [?] analyzed factors leading to project success, identifying governance maturity as critical. The OpenSSF Scorecard [?] operationalized security and maintenance metrics but focuses on security rather than sustainability. O'Mahony and Ferraro [?] studied the emergence of governance in open source communities.

C. Community Dynamics

The CHAOSS project [?] developed standardized community health metrics. Avelino et al. [?] formalized the “truck factor” (bus factor) concept, measuring key-person risk. Constantinou and Mens [?] studied socio-technical evolution in

the Ruby ecosystem. However, these metrics lack validation against actual sustainability outcomes.

D. Ecosystem and Popularity

Borges et al. [?] examined factors driving GitHub popularity, noting stars often reflect marketing rather than quality. Recent work by He et al. [?] revealed millions of fake GitHub stars, challenging popularity-based evaluations.

E. Research Gap

Prior work typically examines single dimensions (governance OR community OR ecosystem). No study has: (1) combined all three dimensions in one predictive framework, (2) validated metrics against ground-truth sustainability labels, or (3) provided interpretable thresholds for practitioners.

III. RESEARCH METHOD

A. Study Design

We conducted a comparative sample study using repository mining, following the ABC framework [?]:

- **Setting:** Neutral (analyzing existing data without manipulation)
- **Strategy:** Sample study examining distribution in population
- **Generalization:** Statistical (sample to GitHub population)

B. Sampling

Table ?? summarizes our sampling strategy.

TABLE I
SAMPLING STRATEGY

Aspect	Value
Source	GHArchive via BigQuery (December 2023)
Total sample	500 projects
Sustainable	250 (active with commits in 2024)
Non-sustainable	250 (archived or dormant ≥ 12 months)
Languages	Python, TypeScript, JavaScript, Go, Java
Minimum stars	500 (ensures non-trivial projects)

C. Data Collection

We collected data in three phases:

Phase 1: Governance metrics. Using the GitHub API, we checked for presence of CODE_OF_CONDUCT, CONTRIBUTING.md, LICENSE, issue templates, PR templates, README, MAINTAINERS.md, GOVERNANCE.md, and CODEOWNERS files.

Phase 2: Community metrics. Using GHArchive, we extracted median issue response time and PR review time. Using the GitHub Contributors API, we calculated bus factor (minimum contributors accounting for 80% of commits) and contributor Gini coefficient.

Phase 3: Ecosystem metrics. Using the GitHub API, we collected stars, forks, and watchers counts.

The final dataset contains 32 features across 500 repositories.

D. Data Analysis

Table ?? summarizes our analysis methods.

TABLE II
ANALYSIS METHODS BY RESEARCH QUESTION

RQ	Method	Justification
RQ1	Chi-square, Fisher's Exact	Categorical variables
RQ2	Mann-Whitney U, Survival	Non-normal distributions
RQ3	Spearman, Decision tree	Non-linear relationships
RQ4	LR, RF, XGBoost, SHAP	Prediction + interpretability

E. Robustness Checks

We applied the following robustness checks:

- **Multiple testing:** FDR correction using Benjamini-Hochberg [?]
- **Confidence intervals:** Bootstrap with 1000 resamples
- **Power analysis:** Confirmed 99% power with n=500

IV. RESULTS

A. RQ1: Governance Practices

Table ?? shows governance adoption by sustainability status.

TABLE III
RQ1: GOVERNANCE ADOPTION BY SUSTAINABILITY STATUS

Practice	Sus.	Non-sus.	χ^2	OR
Contributing guide	57.6%	41.2%	12.8***	1.94
Code of conduct	36.4%	24.8%	7.4**	1.74
PR template	43.6%	19.6%	32.2***	3.17
Maintainer guidelines	18.8%	5.2%	20.6***	4.22

*** $p \leq 0.001$, ** $p \leq 0.01$, * $p \leq 0.05$

Key finding: Maintainer guidelines (MAINTAINERS.md, GOVERNANCE.md, or CODEOWNERS) show the strongest association with sustainability (OR=4.22, $p \leq 0.0001$).

B. RQ2: Community Health

Table ?? shows community metrics by sustainability status.

TABLE IV
RQ2: COMMUNITY METRICS BY SUSTAINABILITY STATUS

Metric	Sus.	Non-sus.	p-value	r
Issue response (days)	0.54	1.78	≤ 0.0001	0.27
Bus factor	4	2	≤ 0.0001	0.30
Contributor Gini	0.86	0.79	≤ 0.0001	0.38

Key finding: Sustainable projects have 2x higher bus factor (median 4 vs. 2), indicating more distributed leadership and lower key-person risk.

C. RQ3: Ecosystem Metrics

Table ?? shows ecosystem thresholds identified through decision tree analysis.

Key finding: Stars correlate with sustainability but with diminishing returns; projects above the threshold are 4.15x more likely to be sustainable.

TABLE V
RQ3: ECOSYSTEM THRESHOLDS

Metric	Threshold	Below	Above	Lift
Stars	11,934	22.4%	92.9%	4.15x
Forks	1,724	31.6%	83.6%	2.65x
Watchers	452	44.4%	88.9%	2.00x

D. RQ4: Combined Prediction

Table ?? shows model performance comparison.

TABLE VI
RQ4: MODEL PERFORMANCE

Model	Accuracy	ROC-AUC
Logistic Regression	86.6% (± 2.3)	0.928 (± 0.017)
Random Forest	95.0% (± 2.3)	0.991 (± 0.004)
XGBoost	97.2% (± 1.2)	0.993 (± 0.007)

SHAP analysis reveals the top predictive features: stars (2.72), total commits (2.25), watchers (1.44), and unique contributors (0.90).

Key finding: Combining governance, community, and ecosystem features achieves 97.2% prediction accuracy, demonstrating that multi-dimensional assessment outperforms single-factor evaluation.

V. DISCUSSION

A. Implications of Findings

Governance maturity matters more than documentation breadth. Maintainer guidelines (OR=4.22) outperform contributing guides (OR=1.94). Maintainers should prioritize explicit governance documentation (MAINTAINERS.md, GOVERNANCE.md) over generic contribution guidelines.

Contributor structure predicts longevity. A bus factor of 4 indicates sustainable leadership distribution. Contributors should assess whether key-person risk exists before joining a project.

Stars are necessary but not sufficient. The threshold effect at approximately 12,000 stars shows diminishing returns. Organizations should not rely solely on star counts for dependency decisions.

Combined models outperform single-dimension evaluations. The 97.2% accuracy demonstrates that multi-factor assessment is superior. All stakeholders should adopt holistic evaluation frameworks.

B. Recommendations for Practitioners

- **Maintainers:** Add MAINTAINERS.md or GOVERNANCE.md early in project lifecycle
- **Contributors:** Check bus factor ≥ 3 and issue response times ≤ 1 day before joining
- **Organizations:** Use combined metrics for dependency evaluation, not just stars
- **Funders:** Prioritize projects with explicit governance documentation

C. Limitations and Threats to Validity

Internal validity. Our binary sustainability classification may oversimplify reality; we mitigated this by testing three alternative definitions (all showed robust results). Additionally, 47% of OpenSSF scores were missing; we addressed this via MICE imputation and sensitivity analysis.

External validity. Results are limited to GitHub projects and may not generalize to GitLab, Bitbucket, self-hosted repositories, or non-software projects. Our 500-star minimum threshold means findings apply to already-visible projects; early-stage or niche projects may exhibit different patterns.

Construct validity. This is an observational study. All findings represent predictive associations, not causal relationships. We cannot claim that adding governance documentation *causes* sustainability—only that it *predicts* it. Correlation may arise from confounding factors (e.g., skilled maintainers implement both governance AND sustain projects).

VI. CONCLUSIONS

We analyzed 500 GitHub projects across governance, community, and ecosystem dimensions to identify factors associated with OSS sustainability. Our main findings are:

- 1) **Maintainer guidelines** show the strongest governance association (OR=4.22, $p < 0.0001$)
- 2) **Bus factor** of 4 differentiates sustainable from non-sustainable projects
- 3) **Stars threshold** of approximately 12,000 provides a meaningful sustainability indicator (4.15x lift)
- 4) **Combined prediction** achieves 97.2% accuracy, demonstrating multi-dimensional assessment superiority

For practitioners: document governance explicitly, build core maintainer teams (bus factor ≥ 4), don't trust stars alone, and use interpretable models for dependency risk assessment.

Future work should examine whether implementing recommended practices causally improves sustainability through longitudinal studies, and extend the analysis to non-GitHub platforms.

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